Junghyeok Lee

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Education

Seoul National University of Science and Technology — Seoul, Korea
Ph.D. student, Department of Electrical and Information Engineering / Advisor: Hyun Kim — Sep. 2026 – present

Seoul National University of Science and Technology — Seoul, Korea
Master of science, Department of Electrical and Information Engineering / Advisor: Hyun Kim — Sep. 2023 – Aug. 2025

Hankuk University of Foreign Studies — Yongin, Korea
Bachelor of science, Department of Electronic Physics — Mar. 2016 – Feb.2023

Experiences

Korea Electronics Technology Institute, KETI — Seongnam, Korea
Researcher, SoC Platform Research Center — Jan. 2026 – Aug. 2026

Seoul National University of Science and Technology — Seoul, Korea
Research intern / Advisor: Hyun Kim — Sep. 2023 – Aug. 2025

Research Interests

- Memory-Centric and Processing-in-Memory Architectures for AI Systems
- Design-Space Exploration Frameworks for Heterogeneous LLM Serving
- Hardware-Aware LLM Compression and Inference Acceleration for On-Device AI

Research Publications

Conference

① **Junghyeok Lee***, JiHoon Jang*, and Hyun Kim, "PriME: PIM-Aware Efficient Compression for Memory-Bound Embedding Layers in sLLMs," *2025 IEEE 43rd International Conference on Computer Design (ICCD)*. IEEE, 2025

② **Junghyeok Lee**, JiHoon Jang, and Hyun Kim, "XNC: XOR and NOT-Based Lossless Compression for Optimizing Unquantized Embedding Layers in Large Language Models," *2025 IEEE International Symposium on Circuits and Systems (ISCAS)*. IEEE, 2025.

Journal

① Inseong Hwang*, **Junghyeok Lee***, Huibeom Kang, Gilhyeon Lee, and Hyun Kim, "Survey of CPU and Memory Simulators in Computer Architecture: A Comprehensive Analysis Including Compiler Integration and Emerging Technology Applications," *Simulation Modelling Practice and Theory (2025)*: 103032.

Domestic Conference

① **Junghyeok Lee**, Byung-Soo Kim, Seokhun Jeon, "AiM-CoSim: A Timing-Functional Co-Simulation Framework for GDDR6-AiM", *Summer Annual Conference of IEIE (2026)*

② Chaelyn Lee, **Junghyeok Lee,** Seokhun Jeon "An Error Model for Characterizing Numerical Error Tolerance in Heterogeneous Accelerator-Based LLM Inference", *Summer Annual Conference of IEIE (2026)*

③ **Junghyeok Lee**, JiHoon Jang, and Hyun Kim, "A Lossless Compression Method for Addressing Quantization Accuracy Drop Caused by Modality Differences in Vision-Language Tokens," *Summer Annual Conference of IEIE (2025): 2852-2854.*

④ **Junghyeok Lee**, JiHoon Jang, and Hyun Kim, "Analysis of Data Compression Ratios for Transformer Models Considering Memory Access Efficiency," *Summer Annual Conference of IEIE (2024): 687-689.*

Patents

① 김현, **이정혁**, 장지훈, “임베딩 레이어의 압축을 통한 메모리를 기반으로 저전력 처리를 위한 전자 장치 및 그 구동 방법 (ELECTRONIC DEVICE FOR LOW-POWER PROCESSING BASED ON MEMORY THROUGH COMPRESSION OF EMBEDDING LAYER AND METHOD OPERATION THEREOF)”, 한국, 서울과학기술대학교 산학협력단, 10-2025-0065237

Awards

① 이길현, 윤석규, 김영찬, **이정혁**, 장지훈, "2023-2024 Student Design Competition Winner in Seoul Chapter", IEEE Circuits and Systems Society, 2024.01.31

Projects

- 2T DRAM 기반 저전력, 고성능 PIM 기술을 위한 회로 및 아키텍처 개발 (Development of circuits and architecture for 2T DRAM-based low-power and high-performance PIM cells), 2022.04~2024.12, 한국연구재단 지원
- 초거대 언어모델을 위한 양자화 기술 및 스케줄링 기법을 포함하는 Transformer 가속 솔루션 개발 (Development of Transformer Acceleration Solution Including Quantization and Scheduling Techniques for LLM), 2024.04~2026.12, 산업통상자원부 지원
- 시뮬레이션 기반 고속/고정확도 데이터센터 워크로드/시스템 분석 플랫폼 개발 (Simulation-based High-speed/High-Accuracy Data Center Workload/System Analysis Platform), 2026.01~2026.09, 정보통신기획평가원(IITP) 지원

Skills

- Verilog
- Python
- C / C++

Scholarship / Supports

- 인공지능 반도체 융합 전문 인력 육성 사업 (Multidisciplinary Research Training and Development Enterprise for AI and Semiconductor Technology), 2020.04~2025.12, 한국연구재단 지원